NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Cloud Computing

Assignment-Week 6

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1:

Fabrication is an attack on:

A) Integrity
B) Confidentiality
C) Authenticity
D) Availability

Correct Answer: C

Detailed Solution: Fabrication is an attack on authenticity.

QUESTION 2:

Reliability and efficiency are basic components of security.

A) True
B) False

Correct Answer: B

Detailed Solution: Confidentiality (Keeping data and resources hidden), Integrity and Availability (Enabling access to data and resources) are the major components of security. Refer to slide-2 of Cloud Security-I.

QUESTION 3:

Spoofing threat on cloud security is an example of.

A) Deception
B) Disruption
C) Usurpation
D) Disclosure

Correct Answer: A, C

Detailed Solution: Spoofing results in deception and usurpation. Refer to slide 6 of Cloud Security - I.

QUESTION 4:

Statement I: Intrusion Detection System (IDS) scans the incoming messages, and creates alerts when suspected scans/attacks are in progress.

Statement II: Authentication is the determination of whether or not an operation is allowed by a certain user.

A) Statement I is TRUE and Statement II is FALSE.
B) Statement II is TRUE and Statement I is FALSE.
C) Both statements are TRUE.
D) Both statements are FALSE.

Correct Answer: A

Detailed Solution: Statement I is correct (Refer slide 24 of Cloud Security-I). Authentication is the identification of legitimate users (Refer slide 18 of Cloud Security-I). Hence, Statement II is false. Correct answer is A**.**

QUESTION 5:

Which of the following is/are hypervisor risks associated with rough hypervisor rootkits?

A) Improper configuration of VM.
B) Hypervisor that hides itself from normal malware detection systems.
C) Hypervisor that creates a covert channel to dump unauthorized code.
D) Vulnerable virtual machine applications like Vmchat, VMftp, Vmcat etc.

Correct Answer: B, C

Detailed Solution: Hypervisor risks associated with rough hypervisor rootkits consist of hypervisors that hide themselves from normal malware detection systems and hypervisors that create a covert channel to dump unauthorized code. Refer slide 20 of Cloud Security - II.

QUESTION 6:

In fault tolerance, replication is mirroring/sharing data over disks which are located in separate physical locations to maintain consistency.

A) True
B) False

Correct Answer: A

Detailed Solution: True. Refer slide 15 of Cloud Security - II.

QUESTION 7:

In which of the following web service(s) based security attacks does(do) the attacker send huge amounts of requests to a certain service, causing denial of service?

A) Injection attack
B) Flooding
C) Cross-site scripting (XSS)
D) Metadata (WSDL) spoofing attack

Correct Answer: B

Detailed Solution: Flooding is the attack where the attacker sends huge amounts of requests to a certain service, causing denial of service. Refer to slide 23 of Cloud Security-II.

QUESTION 8:

When is it better to switch cloud providers?

A) Contract price decrease
B) Provider bankruptcy
C) Provider service shutdown
D) Increase in service quality

Correct Answer: B, C

Detailed Solution: Refer to slide no. 17 of Cloud-Security II.

QUESTION 9:

Which of the following Open-source tools is/are used to perform TCP SYN traceroutes on the Amazon EC2 platform?

A) wget
B) nmap
C) ipconfig
D) hping

Correct Answer: D

Detailed Solution: hping is an open-source tool used to perform TCP SYN traceroutes, which iteratively sends TCP SYN packets with increasing TTLs, until no ACK is received. Refer to slide 12 of Cloud Security III.

QUESTION 10:

In risk-based access control (RAC), computing security uncertainty is addressed.

A) True

B) False

Correct Answer: B

Detailed Solution: In risk-based access control (RAC), computing security uncertainty is not addressed. Hence, the statement is false. Refer to slide 18 of Cloud Security - IV.